

ASSIGNMENT 1

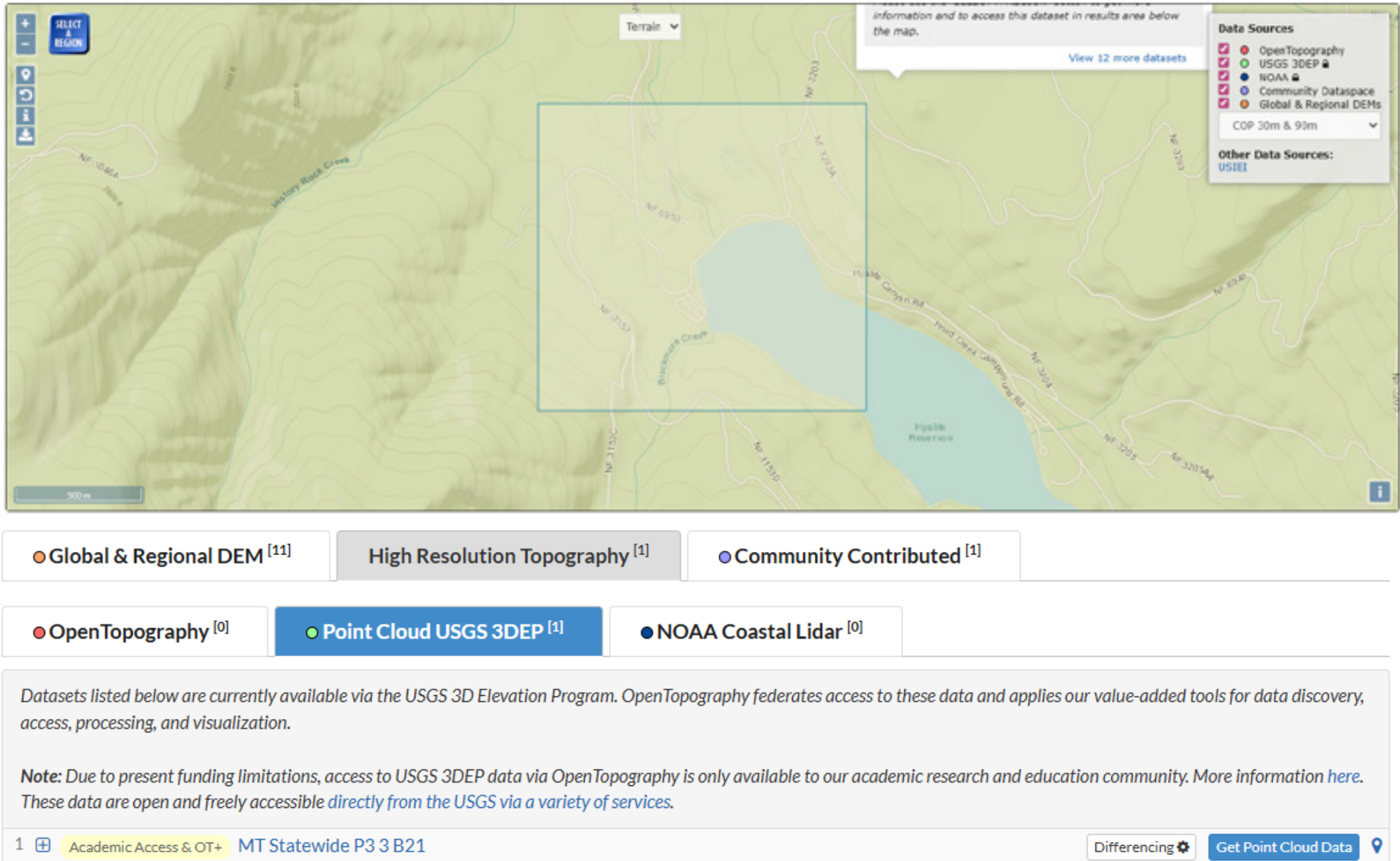
Obtaining LIDAR Data

The first step in creating a topographical model is first gathering topographical data.

In this instance, I used OpenTopography, but other sources such as the Montana LIDAR Inventory also work.

Gathering a point cloud file in the format of a .laz or .las file is preferred. These files are what can be plugged into CloudCompare

I have to remind myself not to get too greedy with the size of the area that I grab. My laptop can't handle as much as I expect it to, and I had to downsize to avoid lag.



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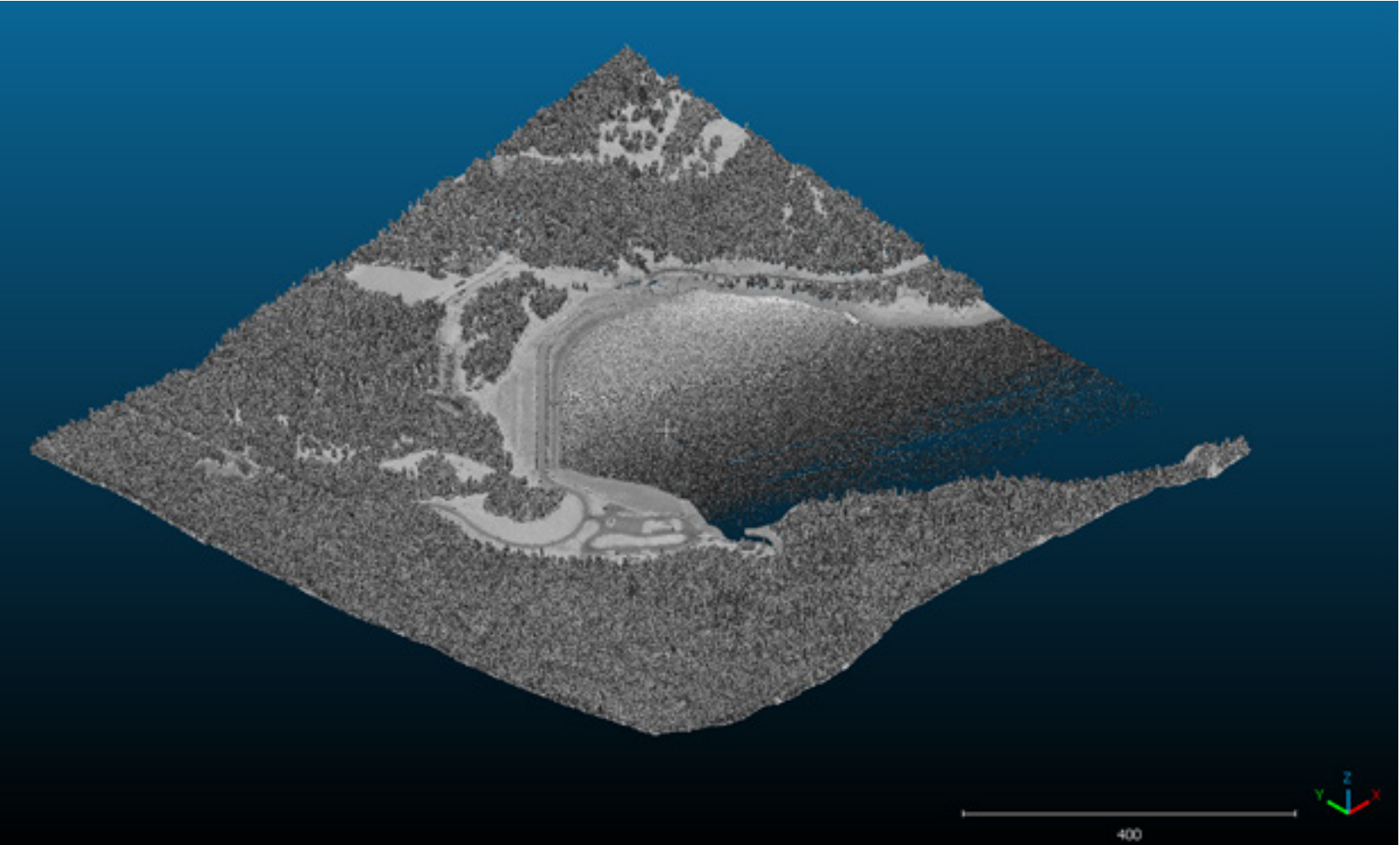
Curating LIDAR Data

The second step is to import the point cloud file into CloudCompare

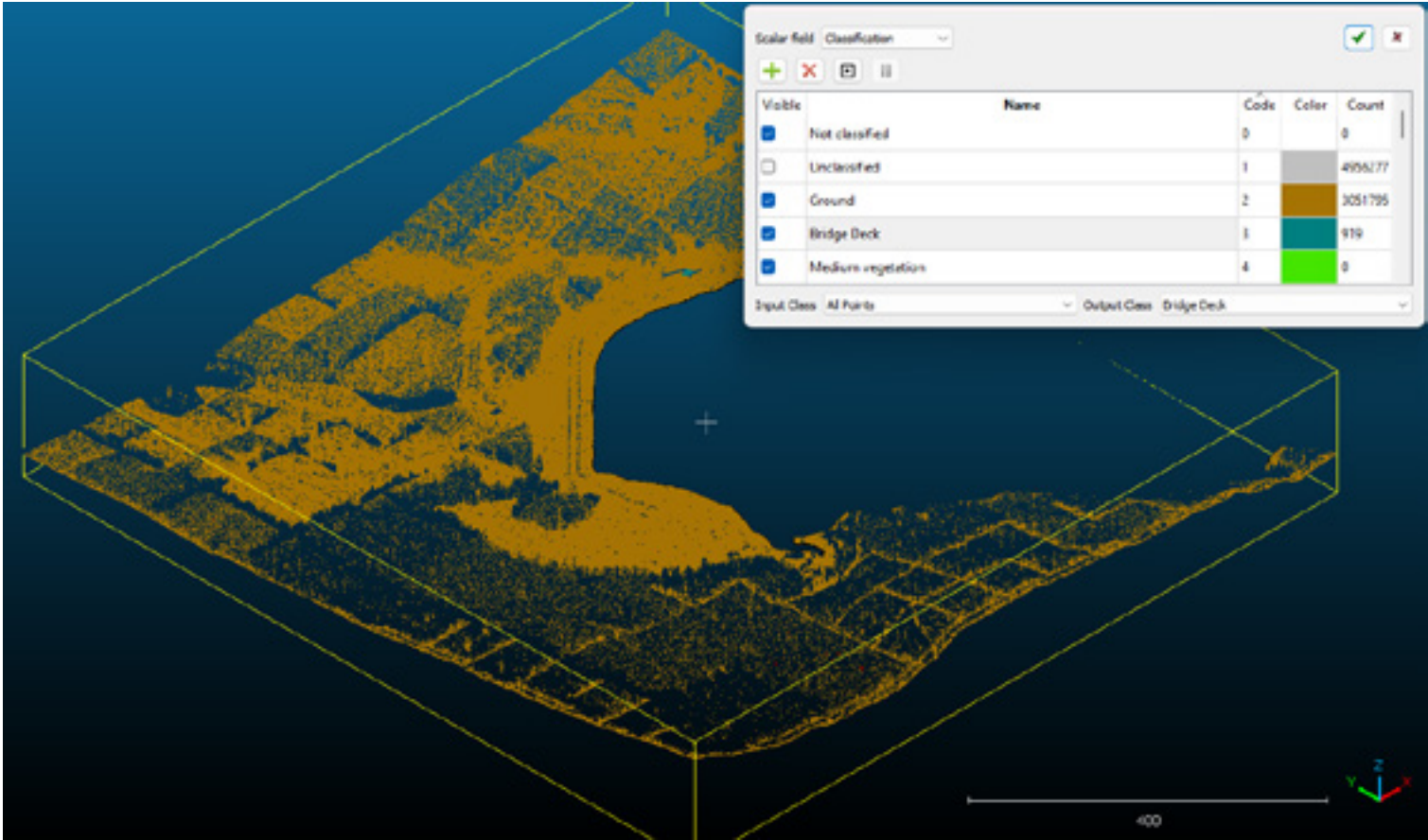
[Fig. 1] This was the original model. I picked the beginning of Hyalite Reservoir.

[Fig. 2] This model in specific had layers applied to it prior to downloading it. To see these layers, I selected “cloud layers” on the right column. Using this feature, I selected only the ground points. I then used “filter points by value” on the top ribbon to isolate them.

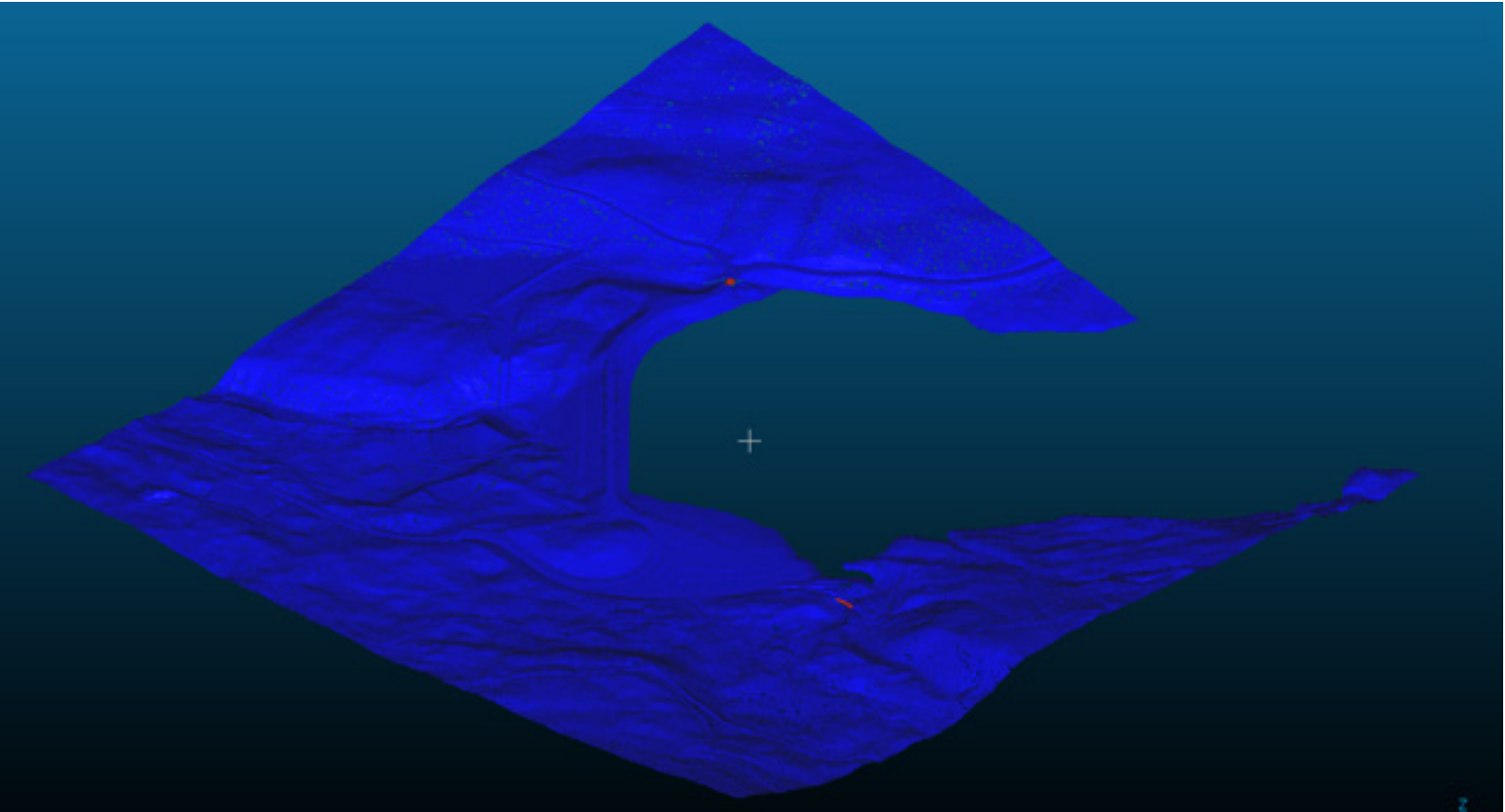
[Fig. 3] Using these points, I selected the mesh operation under edit < mesh. This renders the final mesh that can be imported into Rhino. In this instance, I used a .obj format.



[Fig. 1]



[Fig. 2]



[Fig. 3]

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Rhino Model + Fabrication File

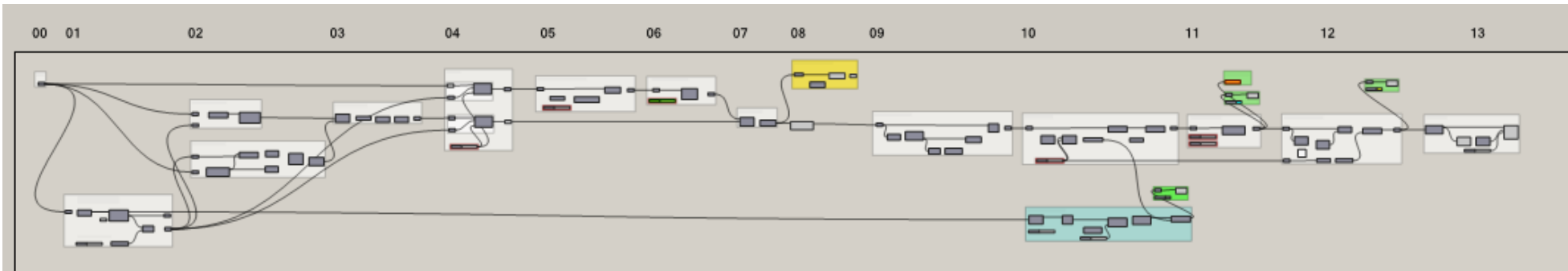
The final step is to produce a fabrication model using Rhino and Grasshopper

[Fig. 1] This shows the grasshopper script used to create the laser cut file. Sections that allow user input are highlighted.

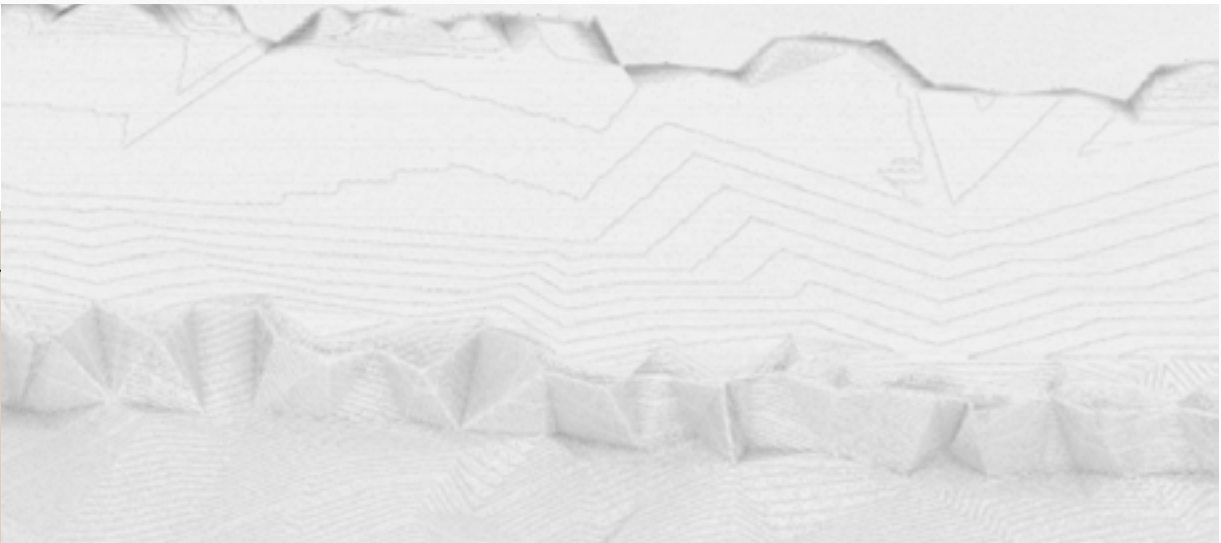
[Fig. 2] Some issues/limitations I noticed when examining my mesh were that inorganic elements (rigid in their shape) such as roads weren't as clean as I would have liked them to be. Maybe there is a possible solution to this that I'm not aware of yet.

[Fig. 3] The layers created by the Grasshopper script are rendered as templates for the laser cutter to read.

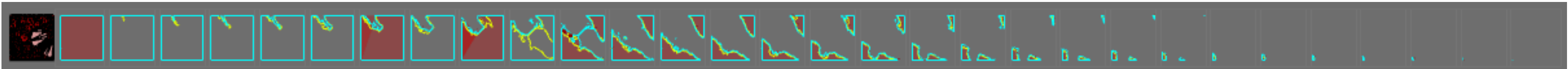
Overall, this entire process significantly lowered the amount of time needed to translate a model to a fabrication-ready instance. It retains much of the actual data of the site, giving a detailed platform from which I can model from.



[Fig. 1]



[Fig. 2]



[Fig. 3]